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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 8

Construct a C program to simulate Round Robin scheduling algorithm with C.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int i, NOP, sum = 0, count = 0;
    int y, quant;
    int wt = 0, tat = 0;
    int at[10], bt[10], temp[10];

    float avg_wt, avg_tat;

    printf("Total number of processes in the system: ");
    scanf("%d", &NOP);

    y = NOP;

    for(i = 0; i < NOP; i++)
    {
        printf("\nEnter the Arrival and Burst time of Process[%d]\n", i +
1);

        printf("Arrival time is: ");
        scanf("%d", &at[i]);

        printf("Burst time is: ");
        scanf("%d", &bt[i]);
```

```
    temp[i] = bt[i];
}

printf("\nEnter the Time Quantum for the process: ");
scanf("%d", &quant);

printf("\nProcess No\tBurst Time\tTAT\tWaiting Time");

sum = 0;
i = 0;

while(y != 0)
{
    if(temp[i] <= quant && temp[i] > 0)
    {
        sum = sum + temp[i];
        temp[i] = 0;
        count = 1;
    }
    else if(temp[i] > 0)
    {
        temp[i] = temp[i] - quant;
        sum = sum + quant;
    }

    if(temp[i] == 0 && count == 1)
    {
        y--;

        printf("\nProcess No[%d]\t%d\t\t%d\t%d",
               i + 1,
               bt[i],
               sum - at[i],
               sum - at[i] - bt[i]);

        wt = wt + sum - at[i] - bt[i];
        tat = tat + sum - at[i];

        count = 0;
    }

    if(i == NOP - 1)
    {
```

```
            i = 0;
        }
        else if(at[i + 1] <= sum)
        {
            i++;
        }
        else
        {
            i = 0;
        }
    }

    avg_wt = (float)wt / NOP;
    avg_tat = (float)tat / NOP;

    printf("\n\nAverage Waiting Time: %.2f", avg_wt);
    printf("\nAverage Turnaround Time: %.2f\n", avg_tat);

    return 0;
}
```

OUTPUT:

